

BUILD ONCE — the contract package (separately metered: \$21.75; Arm B only)

1 · PROVE

spec of record: Contract.lean
1,271 lines · 95 theorems
zero sorry · empty axioms
tier: kernel-evaluated
finite-scope checks (rfl)

2 · TWIN

pure twin implementation
clause-for-clause
transcription
no I/O, no clocks

3 · BRIDGE

conformance suite:
witnesses conform, violating
inputs fail the right clause
samples agreement —
not a refinement proof

4 · ATTACK

adversarial review,
independent reviewers,
distinct lenses; findings
fixed at the causing layer

contract constrains the repair: the fix must name its clause and leave every theorem green

Arm B — FCDD repair session (fresh context; contract package on disk)

PER DEFECT

symptom-level
bug report
(no location,
no cause)

locate violated clause

run bridge on the
faulty build

extend contract if uncovered

+12 to +35 theorems
observed per cell

fix, then re-prove

kernel gate green,
zero sorry

bridge

conformance suite
passes on the fix

attack

independent
reviewers, ≤ 3 rounds

Arm A — ordinary repair session (fresh context; no formal-methods vocabulary)

reproduce symptom

emulator harness

fix

Z80 assembly
edit

add / adjust tests

regression stays
green

review

fresh reviewer,
≤ 3 rounds

SEALED ACCEPTANCE GATE

written before any
repair; hash published

Residuals the method itself states: the contract establishes coherence, never correctness of intent; the bridge samples agreement and cannot catch a spec-twin common-mode error. In this study the contract was built from the pristine engine — an oracle asymmetry Arm A does not share (see the paper's oracle-asymmetry disclosure).